Theta continuity analysis (autocorrelogram peak range)

To measure how continuously theta rhythm was present across behavioural states, we adapted the autocorrelogram (ACG) peak-range method of Dunn et al. (2022). We translated the authors' MATLAB code (`quantify_xcorr_epochs.m`, `create_sine_ref_xcorrs.m`) to Python. Each LFP recording was split into 1-s epochs. Each epoch was given a frequency estimate by matching its autocorrelogram to those of reference sinusoids. The regularity of its oscillation was measured as the normalised height of the first side peak of its autocorrelogram.

LFP import and timing correction

LFP was recorded continuously at $f_{\mathrm{nat}} = 32$ kHz (Neuralynx). Each .ncs record has 512 sample slots, but only the first V_r of them (*NumValidSamples*) contain data. During brief acquisition interruptions the system writes runs of short records ($V_r < 512$). We therefore read only the valid samples of each record. Valid sample j ($j = 0, \dots, V_r - 1$) of record r, which has timestamp T_r (µs), was placed on a uniform 32-kHz grid at index

$$ k_{r,j} = \operatorname{round}\!\left(\frac{(T_r - T_0)\, f_{\mathrm{nat}}}{10^{6}}\right) + j , $$

where T_0 is the timestamp of the first record. Empty grid positions were filled by linear interpolation between the neighbouring recorded samples. Recordings with a gap longer than 1 s were excluded. ADC counts $c[k]$ were converted to microvolts using the *ADBitVolts* value b (V per count) in each file's header:

$$ x[k] = c[k]\, b \cdot 10^{6} \quad (\mu\mathrm{V}). $$

Any epoch that contained a short record was excluded.

ADC saturation

Amplifier clipping was detected on the raw 32-kHz trace before any filtering. Let R be the recording's input range from the file header (*InputRange*: 1,000 or 1,500 µV depending on the recording), and let $\Delta = b \cdot 10^{6}$ µV be one ADC step. A sample was counted as clipped if

$$ |x[k]| \ge R - 1.5\,\Delta . $$

Each run of clipped samples was extended by 200 ms on both sides, because filtering spreads clip edges into neighbouring samples. Any epoch that overlapped an extended run was excluded.

Pre-processing

The following steps were applied to the whole trace, in this order.

1. **Downsampling.** The trace was downsampled from 32 kHz to $f_s = 1$ kHz by polyphase resampling with an anti-aliasing filter.

2. **Line-noise removal.** Second-order IIR notch filters were applied at $f_0 = 50, 100, 150$ and 200 Hz, with quality factor $Q = 30$. Each notch has a −3 dB bandwidth of $\Delta f = f_0 / Q$ (1.7 Hz at 50 Hz). Each filter was applied forward and backward, giving zero phase.

3. **Wavelet denoising.** The trace was decomposed to 4 levels with the *sym4* wavelet, giving approximation coefficients and detail coefficients d_j, $j = 1, \dots, 4$ (level 1 is the finest). The noise level was estimated from the finest level:

   $$\hat\sigma = \frac{\operatorname{median}\left(|d_1|\right)}{0.6745}.$$

   For each level j with M_j coefficients, the signal standard deviation and the BayesShrink threshold (Chang et al., 2000) were

   $$\hat\sigma_{x,j} = \sqrt{\max\!\left(\frac{1}{M_j}\sum_{m} d_{j,m}^{2} - \hat\sigma^{2},\; 0\right)}, \qquad \lambda_j = \frac{\hat\sigma^{2}}{\hat\sigma_{x,j}} .$$

   If $\hat\sigma_{x,j} = 0$, λ_j was set to $\max_m |d_{j,m}|$, which sets that level to zero. Detail coefficients were soft-thresholded,

   $$\eta_{\lambda}(d) = \operatorname{sign}(d)\,\max\left(|d| - \lambda,\; 0\right),$$

   and the trace was reconstructed from the approximation and the thresholded detail coefficients.

4. **Detrending.** The least-squares linear fit $a + bt$ was subtracted from the whole trace.

5. **Outlier replacement.** For each sample n, let m_n and $\mathrm{MAD}_n = \operatorname{median}\left(|x - m_n|\right)$ be the median and median absolute deviation over a 400-ms window (samples $n - 200$ to $n + 199$); the window shrinks at the ends of the trace). A sample was an outlier if

   $$|x[n] - m_n| > 3 \times 1.4826\, \mathrm{MAD}_n .$$

   Outliers were replaced by linear interpolation between the nearest non-outlier samples.

6. **Low-pass filtering.** The trace was low-pass filtered at $f_c = 100$ Hz with a fourth-order Butterworth filter, $|H(f)|^{2} = 1 / \left[1 + (f/f_c)^{2N}\right]$ with $N = 4$. The filter was applied forward and backward, giving zero phase and an overall magnitude response of $|H(f)|^{2}$.

The cleaned trace was divided into non-overlapping 1-s epochs of $N_e = 1{,}000$ samples. Epoch i covers samples iN_e to $(i+1)N_e - 1$. Its start time was taken from the 32-kHz grid as

$$t_i^{\mathrm{start}} = T_0 + \frac{10^{6}}{f_{\mathrm{nat}}}\, \operatorname{round}\!\left(\frac{i N_e f_{\mathrm{nat}}}{f_s}\right) \quad (\mu\mathrm{s}),$$

and its end time is one 1-kHz sample after its last sample.

Epoch rejection

Two further criteria were applied to the cleaned epochs, in addition to the record-integrity and saturation criteria above.

Spectral criterion. For each epoch $x[n]$, $n = 0, \dots, N_e - 1$, we computed a one-sided power spectral density from a single Hann-windowed segment with the mean removed:

$$P(f_k) = \frac{2\left|\sum_{n} w[n]\,\left(x[n] - \bar{x}\right) e^{-i 2\pi k n / N_e}\right|^{2}}{f_s \sum_{n} w[n]^{2}}, \qquad w[n] = \tfrac{1}{2}\left(1 - \cos\frac{2\pi n}{N_e}\right),$$

with frequency resolution $\Delta f = f_s / N_e = 1$ Hz and $f_k = k\,\Delta f$. Band powers were

$$P_{\delta} = \sum_{1 \le f_k \le 3\,\mathrm{Hz}} P(f_k)\, \Delta f , \qquad P_{\theta} = \sum_{3 \le f_k \le 7\,\mathrm{Hz}} P(f_k)\, \Delta f .$$

Epochs with $P_{\delta} > P_{\theta}$ were rejected, because low-frequency artefacts otherwise pass amplitude-based rejection.

Amplitude criterion. For each epoch we computed its peak-to-peak amplitude, $A_i = \max_n x_i[n] - \min_n x_i[n]$. Its robust *z*-score was

$$z_i = \frac{0.6745\,\left(A_i - \tilde{A}\right)}{\operatorname{median}\left(|A - \tilde{A}|\right)},$$

where $\tilde{A}$ is the median amplitude. The median and MAD came only from that recording's epochs that had passed all earlier criteria. Epochs with $z_i > 5$ were rejected.

Movement state

Animal position (x, y) in cm came from the tracking file of each session, which was on the same clock as the LFP timestamps.

Artefact removal. Some steps between consecutive retained frames a and b implied a speed

$$u_{ab} = \frac{\sqrt{(x_b - x_a)^{2} + (y_b - y_a)^{2}}}{t_b - t_a} > 80 \ \mathrm{cm\,s^{-1}} .$$

The later frame of each such step was marked as an artefact. The test was repeated on the remaining frames until no new artefacts were found, and artefact frames were replaced by linear interpolation in time.

Smoothing and speed. The x and y traces were smoothed with a Gaussian kernel, $g_k \propto \exp\left(-k^{2} / 2\sigma^{2}\right)$ with $\sigma = 1$ frame. Running speed at frame j was

$$v_j = \frac{\sqrt{(x_{j+1} - x_j)^{2} + (y_{j+1} - y_j)^{2}}}{t_{j+1} - t_j} .$$

Epoch classification. Each epoch i was given the median speed of the tracking frames within it,

$$\tilde{v}_i = \operatorname{median}\left\{ v_j : t_i^{\mathrm{start}} \le t_j < t_i^{\mathrm{end}} \right\},$$

or the speed of the nearest frame if it contained none. Epochs with $\tilde{v}_i > 4$ cm s^{-1} were classed as moving and epochs with $\tilde{v}_i \le 4$ cm s^{-1} as immobile. Epochs whose speed could not be determined were excluded.

Autocorrelogram

The ACGs were computed from a separate copy of the cleaned trace. This copy was band-pass filtered at 1–20 Hz (fourth-order Butterworth design, which gives an eighth-order band-pass filter, implemented as second-order sections and applied forward and backward). It was split on the same epoch grid. Only epochs that passed all of the criteria above were analysed.

The mean of each epoch was subtracted, $\tilde{x}[n] = x[n] - \bar{x}$, because a DC offset adds a broad triangular component to the ACG that biases matching toward low frequencies. The ACG was computed over all lags and normalised by its zero-lag value:

$$R[\tau] = \sum_{n=0}^{N_e - 1 - |\tau|} \tilde{x}[n]\, \tilde{x}[n + |\tau|], \qquad \rho[\tau] = \frac{R[\tau]}{R[0]}, \qquad \tau = -(N_e - 1), \dots, N_e - 1 ,$$

which gives 1,999 lags (±999 ms) with $\rho[0] = 1$.

Reference matching

A reference bank of 191 sinusoids was synthesised at $f_s = 1$ kHz for one epoch length:

$$s_f[n] = \sin\!\left(\frac{2\pi f n}{f_s}\right), \qquad n = 0, \dots, N_e - 1, \qquad f \in \{1.0, 1.1, \dots, 20.0\}\ \mathrm{Hz}.$$

Their ACGs, $\rho_f[\tau]$, were computed and normalised in the same way. For each epoch we computed the Euclidean distance between its ACG and each reference ACG, divided by the norm of that reference ACG:

$$D(f) = \frac{\left\lVert \rho - \rho_f \right\rVert_2}{\left\lVert \rho_f \right\rVert_2} = \sqrt{\frac{\sum_{\tau}\left(\rho[\tau] - \rho_f[\tau]\right)^{2}}{\sum_{\tau} \rho_f[\tau]^{2}}} .$$

The epoch's frequency estimate and best-match distance were

$$\hat{f} = \arg\min_{f} D(f), \qquad \mathrm{ED}_{\min} = D(\hat{f}) .$$

An epoch was classified as a **theta epoch** if $3 \le \hat{f} \le 7$ Hz, and as **non-theta** otherwise. No threshold was applied to $\mathrm{ED}_{\min}$. The reference bank covers a much wider range than the theta band. As a result, epochs dominated by other rhythms are matched to their own frequency rather than being forced onto the edges of the theta band.

Peak range

Locating the reference features. For each reference ACG ρ_f, local maxima and minima with a prominence of at least 0.05 were identified. We took the first side peak after zero lag, at lag p_f, and the nearest troughs before and after it, at lags $t_{1,f}$ and $t_{2,f}$. Zero crossings of ρ_f were located by linear interpolation. Where ρ_f

changes sign between lags k and $k + 1$, the crossing is at

$$ z = k + \frac{\rho_f[k]}{\rho_f[k] - \rho_f[k+1]} . $$

Search windows. Let $z^{-}(\ell)$ be the last zero crossing before lag ℓ and $z^{+}(\ell)$ the first one after it, each rounded to the nearest lag. The three search windows were

$$ W_P = \left[z^{-}(p_f),\; z^{+}(p_f)\right], \qquad W_{T_1} = \left[z^{-}(t_{1,f}),\; z^{-}(p_f)\right], \qquad W_{T_2} = \left[z^{+}(p_f),\; z^{+}(t_{2,f})\right]. $$

Measuring the epoch. In each epoch's ACG, using the windows of its matched reference $\hat{f}$, we took

$$ P = \max_{\tau \in W_P} \rho[\tau], \qquad T_1 = \min_{\tau \in W_{T_1}} \rho[\tau], \qquad T_2 = \min_{\tau \in W_{T_2}} \rho[\tau]. $$

Peak range and normalised peak range were

$$ \mathrm{PR} = P - \frac{T_1 + T_2}{2}, \qquad \mathrm{PR}_{\mathrm{norm}} = \frac{\mathrm{PR}}{\mathrm{PR}_{\mathrm{ref}}(\hat{f})}, \qquad \mathrm{PR}_{\mathrm{ref}}(f) = \rho_f[p_f] - \frac{\rho_f[t_{1,f}] + \rho_f[t_{2,f}]}{2}. $$

Why the normalisation is needed. Because each epoch has a finite length $T = N_e / f_s = 1$ s, the ACG of a pure sinusoid tapers with lag, approximately $\rho_f(\tau) \approx \left(1 - |\tau|/T\right)\cos(2\pi f \tau)$. Its first side peak is at $\tau \approx 1/f$, with troughs at $1/(2f)$ and $3/(2f)$, so

$$ \mathrm{PR}_{\mathrm{ref}}(f) \approx 2\left(1 - \frac{1}{fT}\right). $$

This is about 1.33 at 3 Hz and 1.71 at 7 Hz. Dividing by $\mathrm{PR}_{\mathrm{ref}}(\hat{f})$ removes this dependence on frequency. $\mathrm{PR}_{\mathrm{norm}}$ is near 1 for a regular, sinusoidal oscillation and near 0 for a flat ACG.

Low-frequency references. The second trough window ends at the zero crossing $\tau \approx 7/(4f)$, which lies inside the epoch only when $f > 1.75$ Hz. References at or below this frequency therefore have no complete search windows. Epochs matched to them (all non-theta) were not assigned a peak range.

Theta continuity measures

For a group of epochs, let n be the number of epochs and k the number classified as non-theta. Subscripts *imm* and *mov* restrict these counts to immobile and moving epochs. We report four measures:

$$ p_{\mathrm{all}} = 100\,\frac{k}{n}, \qquad p_{\mathrm{imm}} = 100\,\frac{k_{\mathrm{imm}}}{n_{\mathrm{imm}}}, \qquad p_{\mathrm{mov}} = 100\,\frac{k_{\mathrm{mov}}}{n_{\mathrm{mov}}}, \qquad q_{\mathrm{imm}} = 100\,\frac{k_{\mathrm{imm}}}{k}. $$

Here q_{imm} is the percentage of non-theta epochs that occurred during immobility.

For the n_g theta epochs of movement state g, the frequency estimate and normalised peak range were summarised as their means (centroids):

$$ \bar{f}_g = \frac{1}{n_g}\sum_{i \in g} \hat{f}_i, \qquad \overline{\mathrm{PR}}_g = \frac{1}{n_g}\sum_{i \in g} \mathrm{PR}_{\mathrm{norm},i}. $$

Normalised peak range was also summarised as the median and interquartile range per animal, channel and movement state.

Across all epochs with both values, the relationship between $\mathrm{ED}_{\min}$ and $\mathrm{PR}_{\mathrm{norm}}$ was assessed with Pearson's correlation,

$$ r = \frac{\sum_i (a_i - \bar{a})(b_i - \bar{b})}{\sqrt{\sum_i (a_i - \bar{a})^{2}}\,\sqrt{\sum_i (b_i - \bar{b})^{2}}}, $$

and with Spearman's ρ_s, which is Pearson's r computed on the ranks of the two variables.

Statistics

Non-theta prevalence was compared across arena types (circular track, linear track, open arena) and across session types within each arena (open arena: control, rotated, zero; linear track: 0°, 90°, 180°, 270°; circular track: standard, no rotation, rotation). We used two levels of analysis.

Epoch level

Each measure was tested on a contingency table with G groups and two categories (non-theta, theta). The table holds observed counts O_{gc}, row totals $O_{g\cdot}$, column totals $O_{\cdot c}$ and grand total n. We used Pearson's chi-square test of independence:

$$ \chi^{2} = \sum_{g,c} \frac{\left(O_{gc} - E_{gc}\right)^{2}}{E_{gc}}, \qquad E_{gc} = \frac{O_{g\cdot}\, O_{\cdot c}}{n}, \qquad \mathrm{df} = (G - 1)(C - 1). $$

Pairwise comparisons. Each pair of groups was compared on its 2 × 2 table with the Yates-corrected statistic,

$$ \chi^{2}_{\mathrm{Yates}} = \sum_{g,c} \frac{\left(|O_{gc} - E_{gc}| - 0.5\right)^{2}}{E_{gc}} . $$

If any expected count was below 5, Fisher's exact test was used instead. Its two-sided *p*-value comes from the hypergeometric distribution of the table with fixed margins. For a 2 × 2 table with cells $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the reported odds ratio is ad / bc.

Immobility vs movement. Within each group, the non-theta rate during immobility was compared with that during movement using the same 2 × 2 test: rows were immobile and moving, columns were non-theta and theta.

Session level

Epochs from the same recording are not independent. We therefore also computed one percentage per recording session s, with the counts of all its channels pooled:

$$ p_s = 100\,\frac{\sum_{\mathrm{ch} \in s} k_{\mathrm{ch}}}{\sum_{\mathrm{ch} \in s} n_{\mathrm{ch}}} . $$

These were compared across groups with the Kruskal–Wallis test. With N sessions in total, n_g sessions and rank sum $\mathcal{R}_g$ in group g, the statistic is

$$ H = \frac{12}{N(N + 1)} \sum_{g} \frac{\mathcal{R}_g^{2}}{n_g} - 3(N + 1), $$

corrected for ties, with $G - 1$ degrees of freedom. Pairs of groups were then compared with two-sided Mann–Whitney *U* tests:

$$ U = \mathcal{R}_1 - \frac{n_1 (n_1 + 1)}{2}, $$

where $\mathcal{R}_1$ is the rank sum of the first group in the pooled ranking.

Multiple comparisons

Pairwise *p*-values were Holm–Bonferroni corrected (Holm, 1979) within each family: the pairwise tests of one measure at one level, or the immobility-vs-movement tests across groups. With the m *p*-values of a family sorted so that $p_{(1)} \le \dots \le p_{(m)}$, the adjusted values were

$$ \tilde{p}_{(i)} = \min\left\{1,\; \max_{j \le i}\, (m - j + 1)\, p_{(j)}\right\}. $$

Omnibus tests were not corrected, and significance was set at $\alpha = 0.05$.

Animal-controlled comparisons

Not every animal was recorded in every arena. The arena comparisons were therefore repeated using only animals recorded in both of the arenas being compared: Fa8477 and Fa5834 for circular track vs open arena, and Fa1059 and Fa23BD for linear track vs open arena.

Software

Analyses were done in Python using NumPy, SciPy, PyWavelets and pandas.

Parameter summary

Step	Parameter	Value
Acquisition	Native sampling rate	32 kHz
Saturation	Clip level; tolerance; margin	±*InputRange* from header; 1.5 ADC steps; ±200 ms
Resampling	Analysis rate f_s	1 kHz
Notch	Frequencies; Q	50, 100, 150, 200 Hz; 30
Wavelet denoising	Wavelet; levels; threshold	sym4; 4; BayesShrink, soft
Outlier fill	Window; threshold	400 ms moving median; 3 scaled MAD
Low-pass	Cut-off; order	100 Hz; 4 (zero phase)
Epochs	Length	1 s (1,000 samples), non-overlapping
Spectral rejection	Bands	reject if $P_{1\text{–}3\,\mathrm{Hz}} > P_{3\text{–}7\,\mathrm{Hz}}$
Amplitude rejection	Robust *z* threshold	5 (peak-to-peak amplitude)

Step	Parameter	Value
Tracking	Jump threshold; smoothing	80 cm s^{-1}; Gaussian σ = 1 frame
Movement	Speed threshold	> 4 cm s^{-1} moving, ≤ 4 cm s^{-1} immobile
ACG input	Band-pass; mean removal	1–20 Hz, Butterworth order 4 (zero phase); per epoch
Reference bank	Range; step	1–20 Hz; 0.1 Hz (191 sinusoids)
Peak detection	Prominence	0.05
Theta acceptance	$\hat{f}$ range	3–7 Hz (inclusive)
Statistics	α; correction	0.05; Holm–Bonferroni

References

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